1.6

Differential equations of the first order in implicit form

1.6.1 Clairaut equations

A **Clairaut equation** is a differential equation of the form

$$y = xy' + \phi(y'), \tag{1.6.1}$$

where $\phi \in C^1(I)$.

We substitute $y' = p$, $p = p(x)$ and by differentiating with respect to x we obtain that

$$p'(x + \phi'(p)) = 0.$$

So, a possible case is $p' = 0$ which becomes

$$y = Cx + \psi(C). \tag{1.6.2}$$

The second case is

$$\begin{cases} x = -\phi'(p) \\ y = -p\phi'(p) + \phi(p) \end{cases}. \tag{1.6.3}$$

The solution (1.6.2) represents the **general solution** of the Clairaut equation, and the parametric solution (1.6.3) is the **singular solution** of the Clairaut equation.

The singular solution (1.6.3) can not be obtained from (1.6.2) by assigning a particular value to the constant C, because the singular solution represents the envelope of the family of straight lines given by the general solution (1.6.2).

☞ **Example 1.6.1** Find the solutions of the differential equation $y = xy' + \sin y'$.

Solution: We substitute $y' = p$, and the equation becomes $y = xp + \sin p$. Differentiating with respect to x we have:

$$p = p + p'x + p'\cos p, \text{ that is } p'(x + \cos p) = 0.$$

The first case is $p' = 0$, that means $p = C$, and we have the general solution

$$y = xC + \sin C, \quad C \in \mathbb{R}.$$

From the second case we have the parametric solution

$$\begin{cases} x = -\cos p \\ y = -p\cos p + \sin p \end{cases}.$$

☑

☞ **Example 1.6.2** Integrate the equation $y + y' \ln y' = xy'$.

Solution: We write the equation in the form $y = xy' - y' \ln y'$.

Substituting $y' = p$ the equation becomes $y = xp - p \ln p$, and differentiating the new equation with respect to x we have

$$p'(x - 1 - \ln p) = 0.$$

The general solution is

$$y = xC - C \ln C, \;\; C \in \mathbb{R}_+,$$

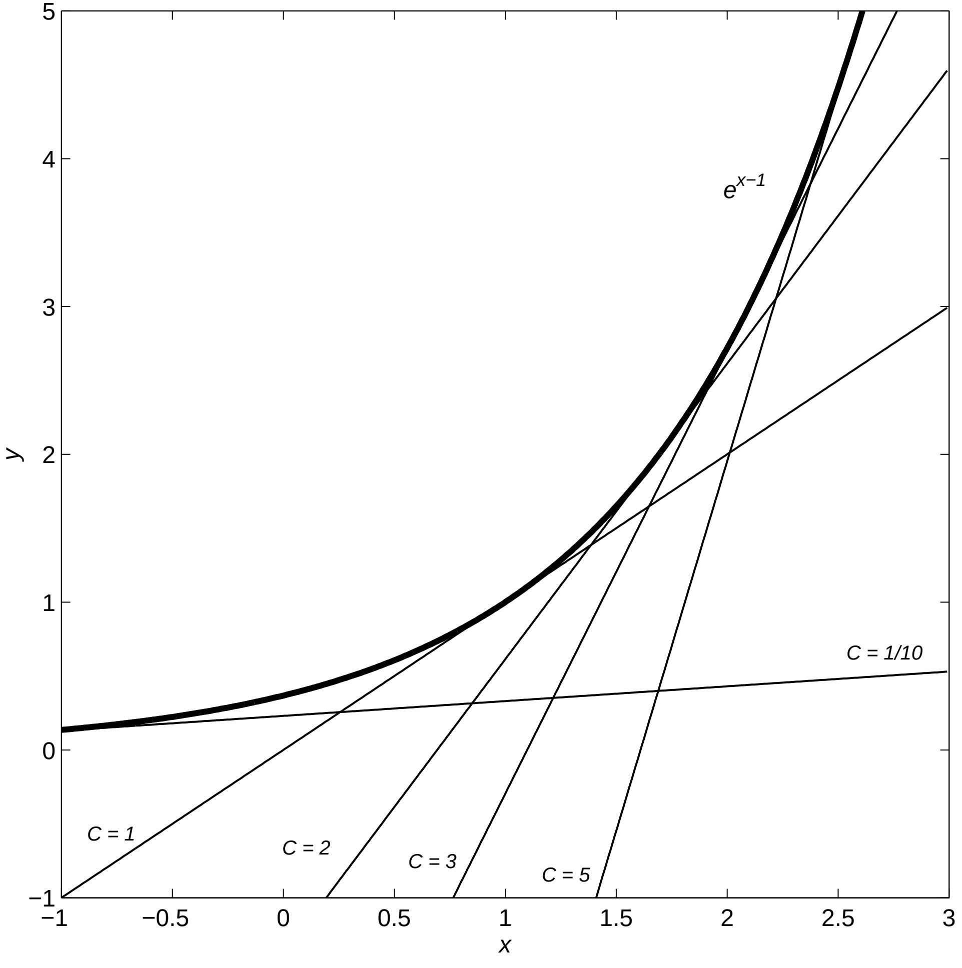
and the singular solution is

$$\begin{cases} x = 1 + \ln p, \;\; p > 0 \\ y = p \end{cases}.$$

Note that the parameter p can be eliminated and the singular integral can be written in the form

$$y = \mathrm{e}^{x-1}.$$

In the following picture we represented a few solutions from the family of straight lines, for some values of the constant C, together with the particular solution. As we can see, the particular solution is indeed the envelope of the



Exercices

1.6.1 Find the solutions of the following differential equations:

i). $y = xy' + \dfrac{a}{y'^2}, a \in \mathbb{R}$

ii). $y = xy' + \dfrac{ay'}{\sqrt{1+y'^2}}, a \in \mathbb{R}$

iii). $y'^3 = 3(xy' - y)$

iv). $yy' = xy'^2 + 1$

v). $2y'^2(y - xy') = 1$

vi). $y = xy' + \mathrm{e}^{-y'}$

vii). $y = xy' - \dfrac{1}{3y'^2}$

viii). $y = xy' + \cos^2 y'$

ix). $y = xy' + a\sqrt{1+y'^2}$

x). $y = xy' - y'^4$

1.6.2 Lagrange Equations

A **Lagrange equation** is an equation of the form

$$y = x\phi(y') + \psi(y'), \tag{1.6.4}$$

with $\phi, \psi \in C^1(I)$.

We substitute $y' = p,\ p = p(x)$, then differentiate the equation (1.6.4) with respect to x and obtain the linear equation of the first order with the unknown x and the independent variable p:

$$\frac{\mathrm{d}x}{\mathrm{d}p} = \frac{\phi'(p)}{p - \phi(p)}x + \frac{\psi'(p)}{p - \phi(p)}.$$

Solving this equation we obtain

$$x = F(p, C),\ C \in \mathbb{R}, \tag{1.6.5}$$

and next

$$y = F(p, C)\phi(p) + \psi(p). \tag{1.6.6}$$

The relations (1.6.5), (1.6.6) represent **the parametric solution** of equation (1.6.4) with the real parameter p.

If $p - \phi(p) = 0$ has the solutions $p_i \in I$, then the equation (1.6.4) has **singular solutions** too, of the form $y = p_i x + \psi(p_i)$, and these solutions are straight lines.

☞ **Example 1.6.3** Find the solutions of the differential equation $y = (1 + y')x + \mathrm{e}^{-y'}$.

➩ **Solution:** Substituting $y' = p$, the equation becomes $y = (1 + p)x + \mathrm{e}^{-p}$ and, differentiating the equation with respect to x, we have $p = 1 + p + xp' - p'\mathrm{e}^{-p}$, which means

$$-1 = p'(x - \mathrm{e}^{-p}),\ p' = \frac{\mathrm{d}p}{\mathrm{d}x}.$$

In what follows we consider the new equation as a differential equation of the first order in x,

$$\frac{\mathrm{d}x}{\mathrm{d}p} + x = \mathrm{e}^{-p} \text{ or } x' + x = \mathrm{e}^{-p}$$

with the solution

$$x(p) = C\mathrm{e}^{-p} + p\mathrm{e}^{-p}. \tag{1.6.7}$$

Next we have

$$y(p) = [(1+p)(C+p)+1]\mathrm{e}^{-p}. \tag{1.6.8}$$

and the relations (1.6.7)+(1.6.8) represent the parametric solution of the problem. In this case we don't have singular solutions. ☑

☞ **Example 1.6.4** Find the solutions of the differential equation $y = 2xy' + \sin y'$.

Solution: Substituting $y' = p$ and differentiating the equation with respect to x, it becomes $-p = p'(2x + \cos p)$. Changing the role of the variables, we get

$$-p\frac{\mathrm{d}x}{\mathrm{d}p} - 2x = \cos p, \text{ or } x' + \frac{2}{p}x = -\frac{\cos p}{p}$$

with the particular solution $\begin{cases} x(p) = \frac{C}{p^2} - \frac{\sin p}{p} - \frac{\cos p}{p^2} \\ y(p) = 2\frac{C}{p} - 2\frac{\cos p}{p} - \sin p \end{cases}$. If $p = 0$, we have a singular solution $y = 0$. ☑

Exercices

✎ **1.6.2** Solve the following differential equations:

i). $y = (1+y')x + \cos(1+y')$

ii). $y = xy'^2 + 1 + y'^2$

iii). $y - 2y'(x - \ln y') = 0$

iv). $y = \frac{3}{2}xy' + \mathrm{e}^{y'}$

v). $y = 2xy' + \ln y'$

vi). $y = xy'^2 + y'^2$

vii). $yy' = 2xy'^2 + 1$

viii). $y = 2xy' + 1 + y'$

ix). $y = (1+y')(1+x)$

x). $y = 2xy' + \sqrt{1+y'^2}$

1.6.3 Implicit equations of the form $f(x, y') = 0$ that can be solved with respect to y or x

☞ **Example 1.6.5** Solve the equation $y = x + y' - \ln y'$.

Solution: We replace $y' = p$, and the equation becomes $y = x + p - \ln p$. Differentiating we obtain that: $y' = 1 + p' - \dfrac{p'}{p}$, which gives $p = 1 + \dfrac{\mathrm{d}p}{\mathrm{d}x} - \dfrac{\mathrm{d}p}{\mathrm{d}x}\dfrac{1}{p}$, and $\mathrm{d}x(p-1) = \mathrm{d}p\left(1 - \dfrac{1}{p}\right)$. Now we have the differential equation $x' = \dfrac{1}{p}$ with the solution $\begin{cases} x = \ln p + C \\ y = p + C \end{cases}$, $C \in \mathbb{R}$. ☑

If we have an equation of the form $f(x, y') = 0$ from which x can be determined explicitly, then we can write it in the form $x = \phi(y')$.

Replacing $y' = p$ we have

$$x = \phi(p) \text{ and } \mathrm{d}x = \phi'(p)\,\mathrm{d}p.$$

From $y' = p$, that is $\dfrac{\mathrm{d}y}{\mathrm{d}x} = p$, we have $\mathrm{d}x = \dfrac{\mathrm{d}y}{p}$ and replacing in the equation we obtain that:

$$\frac{\mathrm{d}y}{p} = \phi'(p)\,\mathrm{d}p, \quad \mathrm{d}y = p\phi'(p)\,\mathrm{d}p.$$

So $\begin{cases} y = \int p\phi'(p)\,\mathrm{d}p + C \\ x = \phi(p) \end{cases}$ that is the solution given in parametric form.

☞ **Example 1.6.6** Solve the differential equation $x = \ln y' + \sin y'$

Solution: We replace $y' = p$, so the equation becomes $x = \ln p + \sin p$. Now we differentiate and obtain: $\mathrm{d}x = \dfrac{\mathrm{d}p}{p} + \cos p\,\mathrm{d}p$, $\dfrac{\mathrm{d}y}{p} = \dfrac{\mathrm{d}p}{p} + \cos p\,\mathrm{d}p$, $\mathrm{d}y = \mathrm{d}p + p\cos p\,\mathrm{d}p$. Next the solution is $\begin{cases} y = p + p\sin p + \cos p + C \\ x = \ln p + \sin p \end{cases}$, with $C \in \mathbb{R}$. ☑

Exercices

✎ **1.6.3** Solve the differential equations:

i). $y = y'^2 + 3xy' + \frac{3}{2}x^2$

ii). $y = -\frac{1}{3}(xy' + x^2 + y'^2)$

iii). $y = y' \ln y'$

iv). $y' = \mathrm{e}^{\frac{y'}{y}}$

v). $x = y'^2 - 2y' + 2$

vi). $x = 2y' + \cos y'$

vii). $x = \mathrm{e}^{y'} + y'$

viii). $x = y'^2 + \sqrt{y'}$

ix). $y = ay'^2 + by'^3$, a, b, constants.

Answers

$\boxed{\mathbf{1.6.1}}$ **i)** $y = xC + \frac{a}{C^2},\ C \in \mathbb{R}^*$; $4y^3 - 27ax^3 = 0$; **ii)** $y = xC + \frac{aC}{\sqrt{1+C^2}},\ C \in \mathbb{R}$; $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$; **iii)** $y = xC - \frac{1}{3}C^3,\ C \in \mathbb{R}$; $9y^2 = \frac{64}{27}x^3$; **iv)** $yC = xC^2 + 1,\ C \in \mathbb{R}$; $y^2 = 4x$; **v)** $2C^2(y - xC) = 1,\ C \in \mathbb{R}$; $8y^3 = 27x^2$; **vi)** $y = xC + \mathrm{e}^{-C},\ C \in \mathbb{R}$; $y = -x\ln|x| + x$; **vii)** $y = xC - \frac{1}{3C^2},\ C \in \mathbb{R}^*$; $4y^3 + 9x^2 = 0$; **viii)** $y = xC + \cos^2 C,\ C \in \mathbb{R}$; $\begin{cases} x = \sin 2p \\ y = p\sin 2p + \cos^2 p \end{cases}$; **ix)** $y = xC + a\sqrt{1 + C^2},\ C \in \mathbb{R}$; $x^2 + y^2 = a^2$; **x)** $y = xC - C^4,\ C \in \mathbb{R}$; $27x^4 - 256y^3 = 0$.

$\boxed{\mathbf{1.6.2}}$ **i)** $\begin{cases} x(p) = C\mathrm{e}^{-p} + \frac{1}{2}[\sin(1+p) - \cos(1+p)] \\ y(p) = C(1+p)\mathrm{e}^{-p} + \frac{1}{2}(1+p)[\sin(1+p) + \cos(1+p)] \end{cases}$;

ii) $\begin{cases} x(p) = \frac{C}{(1-p)^2} + \frac{2p-p^2}{(1-p)^2} \\ y(p) = xp^2 + 1 + p^2 \end{cases}$, $y = 1,\ y = x + 2$; **iii)** $\begin{cases} x(p) = \frac{C}{p^2} + \frac{1}{2} + \ln p \\ y(p) = \frac{2C}{p} + p \end{cases}$;

iv) $\begin{cases} x(p) = \frac{C}{p^3} - \frac{\mathrm{e}^p(2p^2-4p+4)}{p^3} \\ y(p) = \frac{3}{2}xp + \mathrm{e}^p \end{cases}$, $y = 1$; **v)** $\begin{cases} x(p) = \frac{C}{p^2} - \frac{1}{p} \\ y(p) = \ln p + \frac{2C}{p} - 2 \end{cases}$;

vi) $\begin{cases} x(p) = \frac{C}{(p-1)^2} - 1 \\ y(p) = \frac{Cp^2}{(p-1)^2} \end{cases}$, $y = 0,\ y = x + 1$; **vii)** $\begin{cases} x(p) = \frac{1}{p^2}\ln|p| + \frac{C}{p^2} \\ y(p) = 2xp + \frac{1}{p} \end{cases}$;

viii) $\begin{cases} x(p) = \frac{C}{p^2} - \frac{1}{2} \\ y(p) = \frac{2C}{p} + 1 \end{cases}$, $y = 1$; **ix)** $\begin{cases} x(p) = C\mathrm{e}^{-p} - 1 \\ y(p) = (1+p)C\mathrm{e}^{-p} \end{cases}$;

x) $\begin{cases} x(p) = \frac{C}{p^2} - \frac{1}{2p}\sqrt{1+p^2} + \frac{1}{2p^2}\ln\left|p + \sqrt{1+p^2}\right| \\ y(p) = 2xp + \sqrt{1+p^2} \end{cases}$, $y = 1$.

$\boxed{\mathbf{1.6.3}}$ **i)** $y = -\frac{3}{4}x^2$ şi $y = -\frac{1}{2}(-x + C)^2 + \frac{3}{2}C^2,\ C \in \mathbb{R}$;

ii) $y = -\dfrac{x^2}{4}$ şi $y = -\frac{1}{3}\left[x^2 + x(C - 2x) + (C - 2x)^2\right],\ C \in \mathbb{R}$;

iii) $\begin{cases} x = \frac{1}{2}\ln^2 p + \ln p + C \\ y = p\ln p \end{cases}$, $C \in \mathbb{R}$; **iv)** $\begin{cases} x = \ln|\ln p| + \frac{1}{\ln p} + C \\ y = \frac{p}{\ln p} \end{cases}$, $C \in \mathbb{R}$;

v) $\begin{cases} y = \frac{2}{3}p^3 - p^2 + C \\ x = p^2 - 2p + 2 \end{cases}$, $C \in \mathbb{R}$; **vi)** $\begin{cases} y = p^2 + p\cos p - \sin p + C \\ x = 2p + \cos p \end{cases}$, $C \in \mathbb{R}$;

vii) $\begin{cases} y = p\mathrm{e}^p - \mathrm{e}^p + \frac{p^2}{2} + C \\ x = \mathrm{e}^p + p \end{cases}$, $C \in \mathbb{R}$; **viii)** $\begin{cases} y = \frac{2}{3}p^3 + \frac{1}{3}p^{3/2} + C \\ x = p^2 + \sqrt{p} \end{cases}$, $C \in \mathbb{R}$;

ix) $\begin{cases} x = 2ap + \frac{3}{2}bp^2 + C \\ y = ap^2 + bp^3 \end{cases}$, $C \in \mathbb{R}$.